What generalises across recordings? Data leakage, recording-condition confounding and signal-type structure in the Watkins Marine Mammal Sound Database

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Draft v3, 12 October 2026. Sections 3.5–3.7 report analyses that were defined after the first results were seen; they are marked as exploratory in the text.

Abstract

The Watkins Marine Mammal Sound Database is the open archive most often used to develop feature extractors for marine mammal sounds, and recent studies report high species separation on it. We screened all 13,693 clips of the archive, retained 10,694 clips of 36 species from 49 recording sources, and compared five feature extractors (frozen AST, VGGish and EfficientNet-B3; 51 physical descriptors; MFCC) with a baseline that never reads the audio and knows only the sampling rate and duration of a clip. Evaluated as published, with clips divided at random, the extractors separated ten species with a balanced accuracy of 0.64–0.76. With recording sources kept apart the same extractors reached 0.21–0.33, against 0.24 for the baseline. Among the ten nearest neighbours of a clip, 55–82% shared its species, but only 7–13% did so when neighbours were drawn from other sources. Species and source are confounded in the archive: 18 of 36 species come from a single source, and 98% of mysticete clips but 0.1% of odontocete clips were sampled below 16 kHz. What does generalise is coarser. Three signal types defined by time–frequency structure (tonal, impulsive, pulsed) were recognised across sources at 0.48–0.69 within the 40-kHz sampling-rate family, where the baseline fell to chance (0.34). Finer call types were separated by single physical quantities, and taxon followed the frequency band of the signal, with the three taxa overlapping between 1 and 8 kHz. We derive design requirements for an annotation method that reports signal type, measured physical quantities and a band-based taxon probability instead of a species name.

1. Background

Passive acoustic monitoring produces far more audio than analysts can annotate. To organise such recordings we need a representation in which sounds of the same kind lie close together, whichever hydrophone, site or year they come from. The Watkins Marine Mammal Sound Database (Sayigh et al., 2016), with its many species and its annotation of every clip, is the open archive most often used to develop and compare such representations.

Four recent studies report strong separation on it with four different feature extractors. Mel-frequency cepstral coefficients (MFCC) computed after empirical mode decomposition and endpoint detection clustered two to five cetacean species with a mean accuracy of 84.8% (Liang et al., 2025). VGGish embeddings of 0.96-s samples from 12 species, projected to two

dimensions with UMAP, placed mysticetes and odontocetes in distinct regions, and eight of the 12 species formed clear clusters (Cominelli et al., 2024). An Audio Spectrogram Transformer (AST) fine-tuned with metric learning reached 92.4% closed-set species accuracy, and 73.1% without fine-tuning (Lipari et al., 2026). A detector built on a frozen EfficientNet-B3 backbone localised four species in synthetic mixtures with F-scores of 0.82–1.00 (Kim et al., 2026).

Two properties of the archive make these results hard to interpret. First, the clips of a species often come from a few recording events, and consecutive clips were cut from the same tape. When clips are divided at random, or when many short samples are taken from the same files, material from one recording appears on both sides of the comparison. This is a form of data leakage (Kapoor & Narayanan, 2023), and data with such grouping call for a division that keeps the groups apart (Roberts et al., 2017); Lipari et al. (2026) note the possibility themselves. Second, each recording was digitised at a sampling rate chosen for the species concerned, so that recording conditions are confounded with the labels. A classifier can then reach the right answer from the recording conditions and not from the sound, which is known as shortcut learning (Geirhos et al., 2020). Until both effects are measured, it is unclear how much of the reported separation describes the sounds and how much describes the recordings.

A further question is at which level of description acoustic information can be expected to generalise. Operational tools for passive acoustic monitoring are organised by signal type and not by species: PAMGuard, for example, provides a click detector and a whistle and moan detector (Gillespie et al., 2008), and odontocete sounds are conventionally divided into whistles, burst pulses and echolocation clicks (Au & Hastings, 2008). If the species level is confounded in the archive, a coarser description by signal type and frequency band may be the level at which a representation can be validated.

We therefore asked three questions. (i) Do the published extractors separate the labels when evaluated as their authors intended? (ii) How much of that separation remains when recording sources are kept apart, and does it exceed what a baseline that reads no audio achieves from sampling rate and duration alone? (iii) Once recording conditions are controlled, what remains: which label level is recognised across sources, which physical quantities carry it, and how does it relate to taxon? From the answers we derive requirements for a method that annotates marine mammal sounds in new recordings.

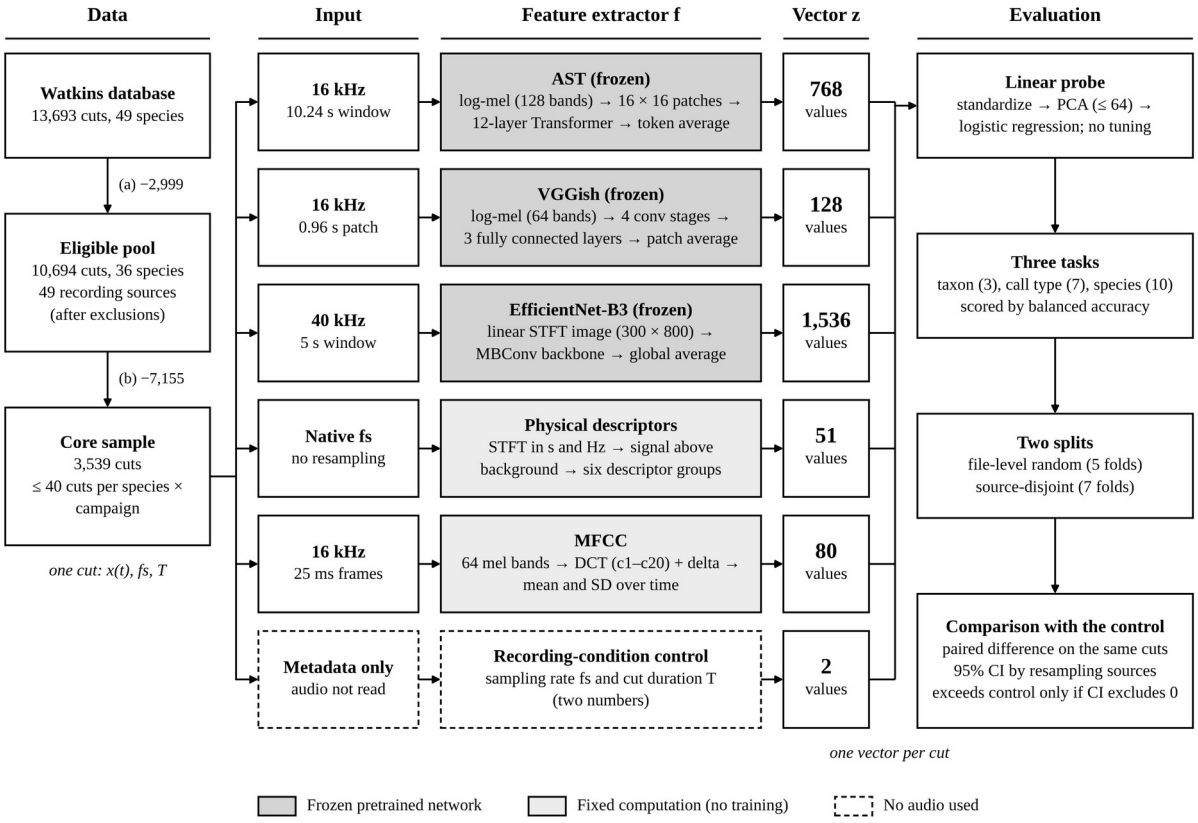
2. Materials and Methods

2.1 Data and screening

We used the Earth Species Project release (v0.1.0) of the Watkins database: 13,693 clips of 49 species at their original sampling rates. Screening removed 2,999 clips. Fin whale clips (567) were excluded for signal quality, clips sampled below 5 kHz (115) were kept as a separate low-frequency pool, and two sea otter clips were dropped. A further 2,315 clips were set aside because their annotation indicated a recording in captivity (2,018), more than one species (233),

a recording in air (114) or a defective recording (9); a clip can carry more than one of these notes. The remaining **eligible pool** contains 10,694 clips of 36 species (8,820 odontocete, 1,385 mysticete and 489 pinniped clips; 5–192 kHz; median duration 1.9 s).

A **recording source** was defined from the metadata as the clips sharing a region and year. The eligible pool has 49 sources. A source is a proxy for a recording campaign and not a verified session or individual. For the comparisons that required all extractors, including EfficientNet-B3, we used the **core sample** drawn earlier from the pool: at most 40 clips per combination of species and field campaign, 3,539 clips of the same 36 species and 49 sources (2,558 odontocete, 585 mysticete, 396 pinniped; 0.25–59 s). Twenty-nine sources (2,769 clips) had been designated as development sources and 20 (770 clips) as held-out sources before any model was fitted. Figure 1 summarises data selection, feature extraction and evaluation.



(a) Set aside before the eligible pool (2,999 cuts): fin whale 567 (signal quality); sampled below 5 kHz 115 (kept as a separate low-frequency pool); sea otter 2; and 2,315 cuts noted as captive 2,018, multi-species 233, in-air 114 or poor quality 9 (a cut can carry more than one of these notes).
(b) Not selected under the cap (7,155 cuts): 5,823 from development sources and 1,332 from test sources.

Figure 1. Data selection, feature extractors and evaluation. Each extractor returns one vector per clip; all vectors are scored with the same linear probe.

2.2 Call-type labels and signal types

The archive annotates each clip with free text. We derived call-type labels from this text by rule: a table of 83 terms maps the words used by the annotators to eight categories (whistle, click, pulse train, moan, pulsed call, song, growl, bark). The words are those of the archive; the eight categories and the mapping were defined in this study. They are weak labels: they describe

a whole clip, 39% of the clips in the core sample name more than one category, and the labels have not been verified by listening. Analyses of call type used the 1,557 core-sample clips whose annotation names exactly one category.

In a second step we regrouped six of the categories into three **signal types** defined by time–frequency structure and not by frequency band or species: *tonal* (whistle and moan; narrow-band, frequency-modulated contours; 852 clips), *impulsive* (click and pulse train; broadband transients that differ in their repetition interval; 407 clips) and *pulsed* (pulsed call and growl; pulse repetition fast enough to produce harmonic bands or a rough sound; 197 clips). Song (86 clips) was excluded because it is a sequence of units and not a signal type, and bark (15 clips) because it was too rare. This regrouping was chosen during the analysis, after the confusion among the eight categories had been inspected, and the results that depend on it (Sections 3.5 –3.7) are exploratory.

2.3 Feature extractors

Table 1 lists what each reference study published and what was applied here; Table 2 gives the input and computation of each extractor. All networks were used with their published weights and were not fine-tuned. Audio was mean-removed and peak-normalised, and at most the first 120 s of a clip were analysed. Each extractor returned one vector per clip. The **recording-condition baseline** used no audio: it consists of the logarithm of the sampling rate and of the clip duration, taken from the metadata.

Table 1. Reference studies: what was published, what was applied here, and what was not reproduced.

Study	As published	Applied here	Not reproduced
Liang et al. (2025)	MFCC after empirical mode decomposition and endpoint detection; k-means and Gaussian-mixture clustering of 2–5 cetacean species. Reported mean accuracy 84.8%.	k-means on clip vectors with k equal to the number of species, for all sets of 2–4 of four species from the study; generic MFCC settings.	Empirical mode decomposition, endpoint detection and the study's MFCC settings. The study was read through a summary of the published text.
Cominelli et al. (2024)	VGGish embeddings of 0.96-s samples (16 kHz) from 12 species (19,682 samples, from files listing a single sound source); two-dimensional UMAP with default parameters, inspected by taxon, species and location. A balanced random forest was applied to a second data set (Placentia Bay), not to Watkins species.	VGGish clip mean (128 values); UMAP of clip vectors; a balanced random forest applied to the Watkins taxon and species labels as a quantitative counterpart of the visual separation.	Sample-level (0.96-s) embeddings, the 12-species selection and the Placentia Bay analyses.
Lipari et al. (2026)	AST pretrained on AudioSet (16 kHz, 10.24-s window), fine-tuned with metric learning; clips divided 70/15/15. Reported species accuracy 92.4%; 73.1% with the frozen network.	Frozen AST (768 values); cosine matching to species mean vectors; clips divided 85/15.	Fine-tuning; open-set and few-shot tests.
Kim et al. (2026)	Frozen EfficientNet-B3 backbone with a YOLO-style detection head; 40-kHz linear spectrogram; four species in synthetic mixtures. Reported F-score 0.82–1.00.	Backbone only: 40-kHz spectrogram in 5-s windows; globally averaged final feature map as clip vector (1,536 values).	Detection head and synthetic mixtures. The clip vector is our construction, so the result concerns the backbone and not the detector.

Table 2. Input and computation of the five feature extractors and the baseline.

Extractor	Sampling rate	Analysis unit	Computation	Values per clip
AST (frozen)[a]	16 kHz	10.24-s window	Log-mel spectrogram (128 bands, to 8 kHz); 16 × 16 patches; 12-layer Transformer; mean of the two class tokens. Short clips zero-padded (first window only) or repeated (all windows averaged).	768
VGGish (frozen)[a]	16 kHz	0.96-s patch	Log-mel spectrogram (64 bands, 125–7,500 Hz); four convolution stages and three fully connected layers; last layer before activation; mean over patches.	128
EfficientNet-B3 (frozen)[a]	40 kHz	5-s window	Linear-frequency spectrogram (0–20 kHz) as a 300 × 800 image; convolutional backbone with ImageNet weights; global average of the final feature map; mean over windows.	1,536
Physical descriptors[b]	native (5–192 kHz)	whole clip	Spectrogram with a 32-ms window; signal defined as power rising above the per-frequency background; six groups: signal band (7), one-third-octave levels (31), temporal structure of events (4), tonality and frequency modulation (3), impulsiveness and click intervals (4), repetition (2).	51
MFCC	16 kHz	whole clip	64 mel bands (50–8,000 Hz), 25-ms frames; coefficients 1–20 and their deltas; mean and standard deviation over frames.	80
Baseline	—	—	No audio. Logarithm of the sampling rate and of the clip duration.	2

[a] Architecture and input follow the defaults of the published model. [b] Defined in this study; thresholds were chosen by the authors.

Three physical descriptors are used individually in Section 3.6. **Tonal concentration** is the share of above-background power that falls within one frequency bin of the strongest bin, taken as the median over the frames in which a signal is present; it is high for a narrow-band contour and low for a broadband transient. The **centre frequency of the signal band** is the frequency below which half of the above-background spectral rise lies. The **inter-click interval** is the median interval between peaks of a 1-ms envelope that exceed eight times its median; it requires at least three peaks and a sampling rate of 8 kHz or more.

2.4 Classification protocols

As published. Each extractor was first evaluated in the manner of the reference studies: a balanced random forest with clips assigned to five folds at random, matching to species mean vectors with a random clip-level split, and k-means clustering scored after optimal assignment. All extractors were passed through every protocol. These analyses used the 2,769 clips of the development sources and were repeated with whole sources assigned to folds.

Common linear probe. To compare the extractors on equal terms we fitted one simple classifier to each of them: median imputation, standardisation, clipping at ±8, at most 64 principal components and class-balanced logistic regression (C = 1), all fitted within the training folds and with nothing tuned. Two divisions were compared: a **random split** of clips into five folds, as in the reference studies, and a **source-disjoint split** in which the 49 sources were assigned

to seven folds, so that no source contributes to both training and testing. A region-disjoint split and the 20 held-out sources served as further checks. Three tasks were scored by balanced accuracy: taxon (3 classes; 3,539 clips), call type (7 categories; 1,542 clips) and species (10 species with enough independent sources; 1,664 clips).

Comparison with the baseline. Every extractor was compared with the recording-condition baseline on the same clips. Confidence intervals (95%) were obtained by resampling recording sources 2,000 times, and an extractor was taken to exceed the baseline only when the interval of the paired difference excluded zero.

Stratification by sampling rate. To remove the sampling rate as a cue, scores were also computed within a sampling-rate family (clips whose rate is nearest to 10, 20, 40, 80 or 160 kHz). Probes were trained on all clips of the training folds and scored on the clips of one family.

Signal types. The three signal types were evaluated with the same probe and the same folds, under the random split, the source-disjoint split, and the source-disjoint split within the 40-kHz and 80-kHz families. To measure what the regrouping discards, the probe was also asked to separate the two categories inside each signal type (whistle from moan, click from pulse train, pulsed call from growl).

2.5 Nearest neighbours and projections

For the 10,694 clips of the eligible pool (four extractors, with AST in two padding variants, and the baseline; EfficientNet-B3 vectors exist only for the core sample) we standardised each representation as above, reduced it to at most 64 principal components and found the ten nearest neighbours of every clip by Euclidean distance. We report how often a neighbour shares the label of the clip, once with neighbours drawn from all clips and once with neighbours drawn only from other recording sources, against the share expected from a random draw. Thirteen very short clips for which no physical descriptor could be computed were omitted for that representation.

To examine how a projection that is given the labels behaves, we fitted UMAP (McInnes et al., 2018; 30 neighbours, minimum distance 0.1) to the core sample in four ways: without labels; with the species label of every clip supplied as a categorical target (target weight 0.5); with labels supplied for the 29 development sources only, the clips of the 20 held-out sources then being projected into the fitted map without their labels; and with species labels permuted at random among the clips. Separation in the two-dimensional map was measured as the balanced accuracy with which the species of a clip is recovered from the majority of its 15 nearest neighbours. Cominelli et al. (2024) used UMAP without labels on 0.96-s samples; the label-supplied variant is our own stress test and not a replication of their analysis. One random seed was used.

2.6 Signal band and taxon

Each clip with a signal-type label was assigned to one of three bands by the centre frequency of its signal band (below 1 kHz, 1–8 kHz, above 8 kHz), and the taxon composition of each combination of signal type and band was tabulated. The centre frequency was measured at the native sampling rate and can therefore not exceed half of that rate.

3. Results

3.1 Species, recording source and sampling rate are confounded

Of the 36 species in the eligible pool, 18 come from a single recording source and seven from two (Figure 2a). Forty-five of the 49 sources contain a single taxon and 38 a single species, and the largest source holds 36% of all clips. For half of the species the archive therefore cannot distinguish the sound of the species from the sound of one recording. Sampling rate is tied to taxon in the same way (Figure 2b): 98% of mysticete clips were sampled below 16 kHz, against 0.1% of odontocete clips and 28% of pinniped clips.

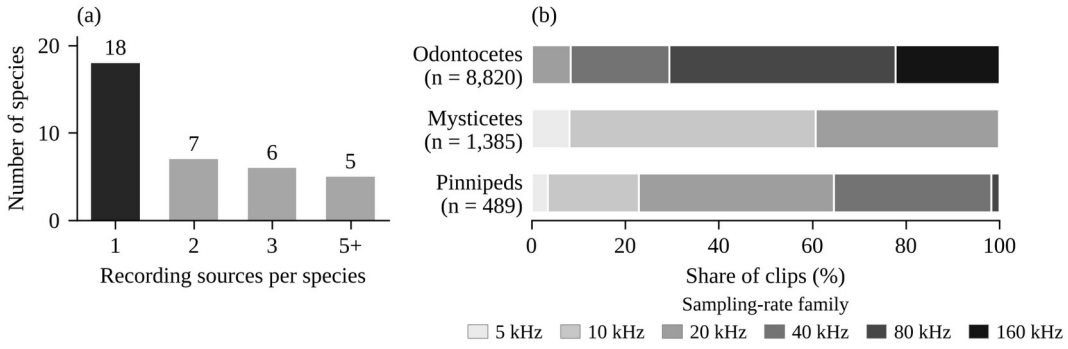


Figure 2. Structure of the eligible pool (10,694 clips, 36 species, 49 recording sources). (a) Number of species by the number of recording sources they come from. (b) Share of clips in each sampling-rate family, by taxon.

3.2 Evaluated as published, the extractors separate the labels; with sources kept apart, most of the separation disappears

With clips divided at random, every protocol reproduced the kind of separation its study reports (Table 3). A balanced random forest on VGGish vectors reached 0.79 for taxon and 0.67 for 27 species. Matching frozen AST vectors to species means identified 0.58 of clips among 29 species (0.73 in Lipari et al., on their selection of clips), and k-means recovered sets of two to four species with an accuracy of 0.66 from MFCC and 0.83 from AST (0.85 in Liang et al., with their denoising and detection). The low scores reported below are therefore not the result of a faulty implementation.

With recording sources kept apart, the random forest no longer recognised species: balanced accuracy fell to 0.02–0.06 for every extractor. Prototype matching fell from 0.58 to 0.12 for AST, and taxon accuracy fell to 0.49–0.63.

Table 3. Evaluation in the manner of the reference studies, with clips divided at random and with recording sources kept apart (development sources, 2,769 clips). Balanced accuracy for the random forest, accuracy for prototype matching, clustering accuracy for k-means. No confidence intervals were computed for this table.

Protocol	Division	Baseline	MFCC	Physical	VGGish	EffNet-B3	AST
Balanced random forest, taxon	random	0.94	0.83	0.88	0.79	0.90	0.81
	sources apart	0.72	0.56	0.61	0.49	0.63	0.56
Balanced random forest, 27 species	random	0.52	0.80	0.69	0.67	0.72	0.69
	sources apart	0.11	0.02	0.04	0.03	0.06	0.04
Prototype matching, 29 species (Lipari et al.)	random	0.10	0.41	0.30	0.40	0.53	0.58
	sources apart	0.10	0.07	0.08	0.07	0.14	0.12
k-means, sets of 2–4 species (Liang et al.)	none	0.70	0.66	0.56	0.65	0.78	0.83

The common linear probe over the core sample showed the same pattern (Figure 3). For ten species, balanced accuracy was 0.64–0.76 under the random split and 0.21–0.33 with sources kept apart (chance 0.10). For the seven call-type categories it fell from 0.55–0.74 to 0.29–0.48, and for taxon from 0.82–0.93 to 0.63–0.80. The loss grows with the fineness of the label: about 0.4 for species, 0.3 for call type and 0.2 for taxon.

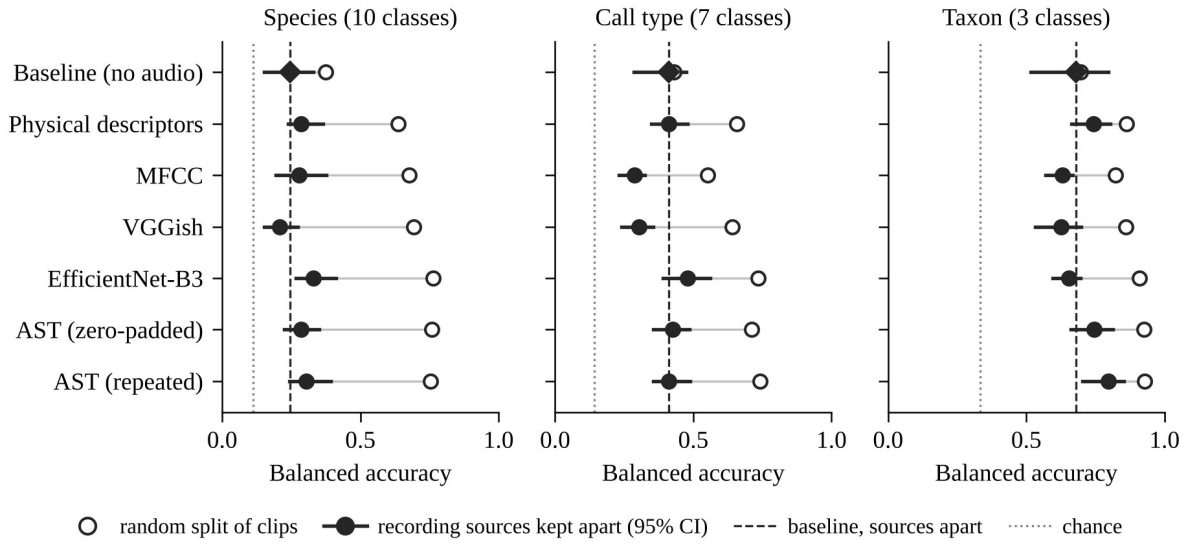


Figure 3. Balanced accuracy of the linear probe for each feature extractor (core sample, 3,539 clips). Open circles: random split of clips, as in the reference studies. Filled circles: recording sources kept apart, with 95% confidence intervals. The dashed line marks the recording-condition baseline with sources kept apart; the dotted line marks chance.

3.3 A baseline that reads no audio does as well

From sampling rate and duration alone, with sources kept apart, the baseline reached 0.24 for species, 0.41 for call type and 0.68 for taxon (Figure 3, dashed lines). Under the published-style protocols it was the best performer for taxon (0.94 with a random division; Table 3). In the source-disjoint cross-validation no extractor exceeded the baseline on any of the three tasks: the largest difference was that of AST for taxon, +0.12 (95% interval −0.01 to +0.26). AST did

exceed the baseline for taxon under the region-disjoint split (+0.13; +0.04 to +0.25) and on the held-out sources (+0.11; +0.01 to +0.26), and the physical descriptors did so for species on the held-out sources only (+0.16; +0.02 to +0.37). For call type no extractor exceeded the baseline under any division, and MFCC was lower than the baseline (−0.12; −0.20 to −0.01).

Individual classes show how the baseline succeeds. It recognised 0.73 of bowhead whale clips, more than AST (0.62), and 0.72 of song clips, which are long (median 6.8 s against 0.7–2.1 s for the other categories), where AST recognised almost none. A score at this level therefore does not show that an extractor has captured the sound.

3.4 Embedding neighbourhoods follow the recording source, and a projection given the labels does not transfer to new sources

In the eligible pool, 55–82% of the ten nearest neighbours of a clip shared its species, far above the 6% expected by chance (Table 4). The same neighbours came from the same recording source in 61–85% of cases. When neighbours were drawn only from other sources, species agreement fell to 7–13% (chance 3%), which is the level of the baseline (10%). Taxon agreement among neighbours from other sources was 0.80–0.88 for the extractors and 0.90 for the baseline. For call type, four of the five representations were slightly above the baseline (0.42 –0.44 against 0.38; MFCC 0.36).

Table 4. Share of the ten nearest neighbours that carry the same label as the clip (eligible pool, 10,694 clips), with neighbours drawn from all clips and from other recording sources only. Call type uses the clips with a single weak label.

Extractor	Species all	Species other sources	Taxon all	Taxon other sources	Call type all	Call type other sources	Same source
Physical descriptors	0.55	0.07	0.93	0.85	0.71	0.43	0.61
MFCC	0.71	0.13	0.96	0.83	0.75	0.36	0.75
VGGish	0.73	0.07	0.95	0.80	0.76	0.42	0.76
AST (zero-padded)	0.82	0.10	0.98	0.87	0.82	0.44	0.85
AST (repeated)	0.82	0.09	0.98	0.88	0.82	0.44	0.84
Baseline (no audio)	0.45	0.10	0.96	0.90	0.59	0.38	0.54
Expected by chance	0.06	0.03	0.70	0.67	0.26	0.24	0.18

A projection that is given the labels makes the same point visually (Figure 4). Without labels, the UMAP of the VGGish vectors recovered species from neighbours with a balanced accuracy of 0.44. With the species labels supplied, separation rose to 0.83, and the map shows compact, well-separated islands. When the map was fitted on the development sources and clips from the 20 held-out sources were projected into it, the fitted clips were separated at 0.77 but the held-out clips at 0.16 (chance 0.06), close to the 0.12 obtained by projecting them into a map fitted without labels. Across the six representations the label-supplied map separated species at 0.80 –0.90 and the held-out clips at 0.13–0.18.

With permuted labels the same procedure did not produce separation (0.06–0.10, with 36 classes). The islands of Figure 4b are therefore not an artefact of the algorithm: within the clips

used to fit the map, the species do differ. That difference, however, is tied to the recording sources in the fit and does not carry over to new sources.

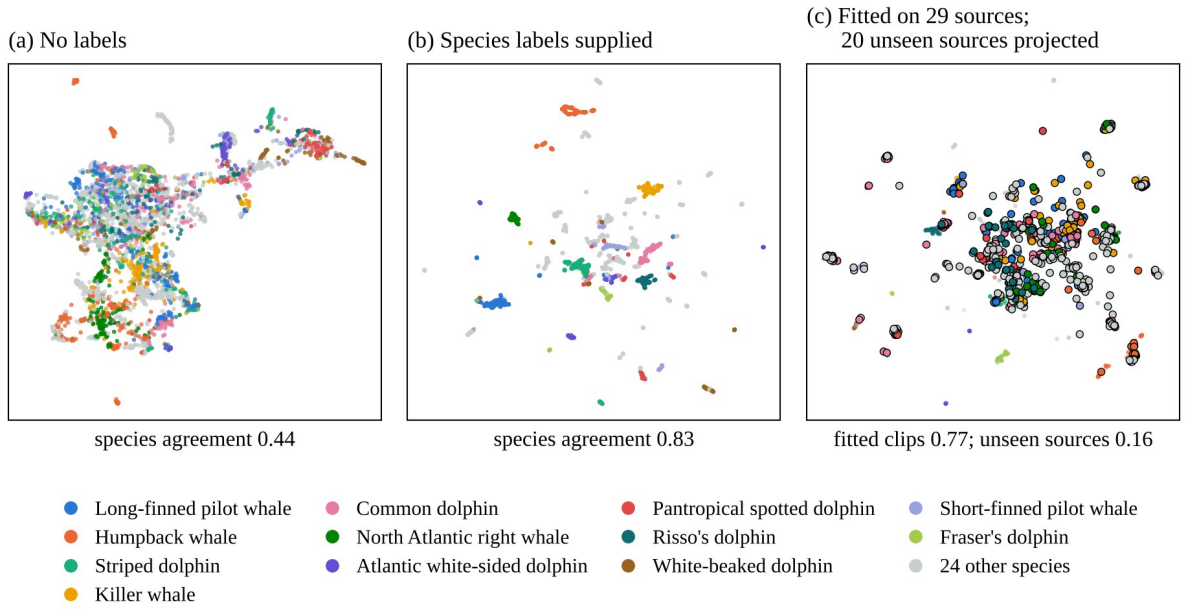


Figure 4. UMAP of VGGish clip vectors (core sample, 3,539 clips), coloured by the 12 most frequent species. (a) Fitted without labels. (b) Fitted with the species label of every clip. (c) Fitted with labels on the 29 development sources (faint points); clips of the 20 held-out sources were projected without labels (outlined points). Numbers give the balanced accuracy with which species is recovered from the 15 nearest neighbours in the map. Axes have no physical meaning. One random seed.

3.5 Three signal types are recognised across sources once the sampling rate is held fixed (exploratory)

Regrouped into three signal types, the labels lost little under the source-disjoint split (Figure 5a, b). EfficientNet-B3 reached 0.82 with a random split and 0.70 with sources kept apart, AST 0.79 and 0.63–0.65, the physical descriptors 0.76 and 0.61; the loss of 0.12–0.16 compares with about 0.4 for species. Over all clips, however, the baseline was high (0.56), because signal type is also tied to sampling rate, and no extractor exceeded it; the largest difference was +0.14 for EfficientNet-B3 (−0.02 to +0.26).

Within one sampling-rate family the baseline has no information left. In the 40-kHz family (368 clips from 19 sources, containing all three types) it fell to chance (0.34), and all six representations (the five extractors, with AST in two padding variants) exceeded it (Figure 5c): EfficientNet-B3 by +0.35 (+0.20 to +0.43), AST by +0.27 (+0.07 to +0.44 across the two padding variants), the physical descriptors by +0.21 (+0.12 to +0.33), MFCC by +0.19 (+0.11 to +0.32) and VGGish by +0.14 (+0.06 to +0.33). In the 80-kHz family (581 clips, 13 sources) three extractors exceeded the baseline (EfficientNet-B3 +0.27, zero-padded AST +0.22, physical descriptors +0.18); this family contains only 12 pulsed clips, so the comparison there is in effect between tonal and impulsive signals. In the 20-kHz family (177 clips, 14 sources) the differences were of similar size (+0.13 to +0.15 for the best extractors) but their intervals included zero.

The pulsed type was the weakest: recall across sources was 0.42–0.56 for the three best extractors, against 0.55–0.70 for tonal and 0.73–0.87 for impulsive signals. For comparison, with the original seven categories the largest difference from the baseline over all clips was +0.06 (−0.02 to +0.17).

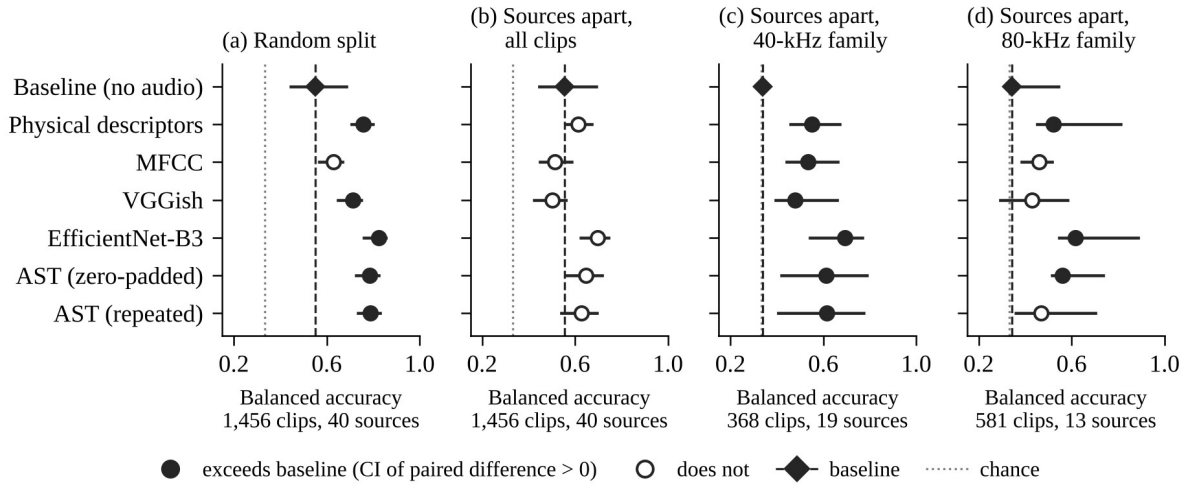


Figure 5. Balanced accuracy for three signal types (tonal, impulsive, pulsed; 1,456 clips with a single weak label). (a) Random split. (b) Recording sources kept apart. (c, d) Sources kept apart, scored within the 40-kHz and 80-kHz sampling-rate families. Bars show 95% confidence intervals; filled circles mark extractors whose paired difference from the baseline excludes zero.

3.6 Finer call types correspond to single physical quantities (exploratory)

Merging categories raises the question of what is lost. When the probe was asked to separate the two categories inside each signal type, with sources kept apart, no extractor exceeded the baseline: whistle against moan gave 0.92 for the baseline and 0.71–0.95 for the extractors, click against pulse train 0.72 and 0.54–0.74, pulsed call against growl 0.72 and 0.57–0.71. In this archive the finer distinctions are thus already predicted by the recording conditions, and no additional acoustic information could be demonstrated for them.

The distinctions themselves are nevertheless physical, and each is carried by one measured quantity (Figure 6). Tonal concentration alone separated tonal from impulsive clips with an area under the ROC curve (AUROC) of 0.95, and of 0.93–0.95 within the 40-kHz and 80-kHz families. Pulsed clips overlap with tonal clips on this quantity, as expected for signals whose fast pulse repetition appears as harmonic bands. Whistles and moans differed in the centre frequency of the signal band (medians 7.8 kHz and 0.9 kHz; AUROC 0.90), although sampling rate alone separates them even better (0.96), so that this distinction remains confounded. Clicks and pulse trains differed in the inter-click interval (medians 11 ms and 3.5 ms; AUROC 0.76, and 0.81 within the 80-kHz family), a distinction that sampling rate alone does not make (0.58). Pulsed calls and growls differed in the lower edge of the signal band (medians 710 Hz and 200 Hz; AUROC 0.79). These values involve no training and are therefore free of leakage, but they rest on weak labels.

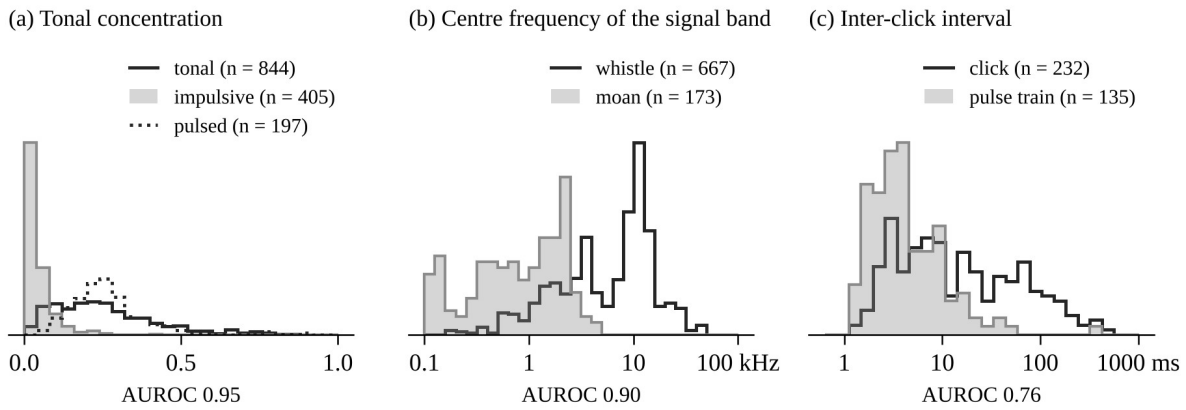


Figure 6. Distributions of single physical descriptors (core sample; no classifier). (a) Tonal concentration for the three signal types. (b) Centre frequency of the signal band for whistles and moans. (c) Median inter-click interval for clicks and pulse trains. AUROC refers to the first two groups of each panel.

3.7 Taxon follows the frequency band of the signal, and the three taxa overlap between 1 and 8 kHz (exploratory)

Signal type alone says little about taxon. Impulsive clips were 95% odontocete, but pulsed clips were 48% mysticete and 45% odontocete, and tonal clips 62% odontocete, 25% mysticete and 12% pinniped. Assigning each signal type to its most frequent taxon gives a balanced accuracy of 0.41 (chance 0.33). The 95% for impulsive signals is a property of this archive: 21 impulsive clips are from pinnipeds, and mysticetes, which also produce impulsive sounds, contribute a single clip.

Taxon follows the frequency band (Figure 7). Above 8 kHz, clips of every signal type were 98–100% odontocete. Below 1 kHz, tonal and pulsed clips were 85–87% mysticete. Between 1 and 8 kHz the taxa overlap: tonal clips there were 51% odontocete, 26% mysticete and 24% pinniped, and pulsed clips 53% odontocete and 40% mysticete. Four in five pinniped clips (112 of 138) lie in this band, where pinnipeds are never the majority. Signal type and taxon thus answer different questions, what kind of signal and who produced it, and it is the band and not the type that links to taxon. The band from 1 to 8 kHz is where a classifier of taxon has real work to do.

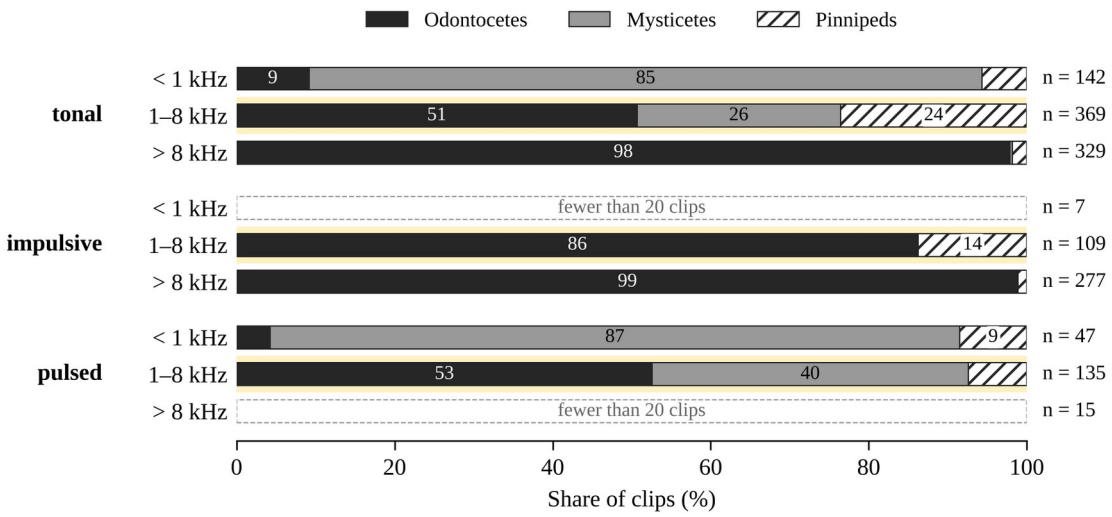


Figure 7. Taxon composition of each combination of signal type and frequency band (1,430 clips with a measurable signal band). Numbers inside the bars are percentages. The band from 1 to 8 kHz, where the three taxa overlap, is shaded.

4. Discussion

4.1 What the published scores measure

Evaluated as their authors intended, the reference extractors separate marine mammal sounds in the Watkins archive, and our values are close to the published ones. The same evaluations, however, give equal or higher scores to two numbers that describe how a clip was digitised, and species separation is lost almost entirely once recording sources are kept apart. Three independent views agree: classification under two divisions (Figure 3), nearest neighbours restricted to other sources (Table 4), and the projection of unseen sources into a map fitted with labels (Figure 4). On this archive a random division of clips measures mainly whether a recording can be recognised again. This is a property of the archive and of the division, and not a defect of any one extractor.

For the user of such a system the consequence is concrete. A classifier reported at 0.7 for species would be expected to perform at 0.2–0.3 on a new deployment, and a classifier that ignored the audio and looked only at the recorder settings would do about as well.

4.2 What generalises: signal type and frequency band

The extractors do carry information about the sounds. It becomes visible where the recording conditions cannot supply the answer, and it lies at a coarser level than species. Three signal types defined by time–frequency structure were recognised across sources well above a baseline that had fallen to chance, by every extractor in the 40-kHz family and by the 51 physical descriptors in both families examined. The loss from leakage at this level was a third of that for species. Taxon, in turn, was linked to the frequency band of the signal and not to its type, with two easy ends and a band of real overlap between 1 and 8 kHz.

This agrees with how passive acoustic monitoring is practised. Detectors are built for signal types, and species or taxon is decided afterwards within a type from frequency and timing. Our results suggest that the Watkins archive supports the validation of the first of these steps, and of a band-based prior for taxon, but not of a species classifier.

4.3 Call-type labels and the limits of fine categories

The call-type labels deserve a separate caution. The words come from annotations written over decades by different people; the eight categories are ours, and 434 of the 1,557 single labels rest on an interpretation of a word (for example *chirp* read as whistle). The categories are also concentrated in species: 78% of moan clips are from the North Atlantic right whale, song comes from three species and three sources, and 93% of bark clips from one species. A fine call-type label is thus often a species label under another name, and inherits the confound. Consistent with this, no extractor separated the categories within a signal type better than the recording conditions did.

We do not conclude that fine call types are unimportant. The distinctions are real and each corresponds to a quantity that an analyst would use: centre frequency, inter-click interval, lower band edge. The conclusion is that, in this archive, they cannot be validated as classes, and that they are better reported as measured values than predicted as categories. Song is a different kind of object, a sequence of units, and belongs to a level above the signal type.

4.4 Other readings of the results

Several alternative readings have to be kept open. A recording source is also a population, a place and a behavioural context, so structure that follows sources need not be technical; the metadata cannot separate the two. Sampling rate is not pure nuisance either: it was chosen to suit the frequency range of the species, and so carries biological information. It is, however, information that is not available when all species are recorded with one instrument, which is the situation in monitoring. The low scores of the networks pretrained at 16 kHz may reflect their input and not their representation: in 22% of the core sample the signal band lies entirely above 8 kHz and is removed by resampling. Finally, the absence of species generalisation is not evidence that species cannot be classified. Half of the species cannot be tested across sources at all, only frozen networks with a linear probe were examined, and for some species the audio clearly helped (bottlenose dolphin recall was 0.00 for the baseline and 0.35 for AST).

4.5 Design requirements for an annotation method

Table 5 turns the observations into requirements. Together they describe a method in three stages: find the intervals in which a signal rises above the background, at the native sampling rate; measure physical quantities for each interval in seconds and hertz; and report the signal type, the measured values and a taxon probability based on the band. Finer names (whistle-like, burst-like) would be attached by rule to the measured values, so that the reason for each label

is visible in the output. No species name is reported. The stages are a proposal; none has been implemented or tested as a whole.

Table 5. Design requirements derived from the screening results. The proposed elements were not tested in this study.

Observation	Requirement	Proposed element
Species accuracy fell from 0.64–0.76 to 0.21 –0.33 when sources were kept apart; 18 of 36 species come from one source.	Do not make the species name the target; evaluate with recording sources kept apart.	Output without species; source-disjoint evaluation as the default.
Sampling rate and duration alone gave 0.24 for species and 0.68 for taxon.	Report every result next to a baseline that reads no audio, and confirm it with the sampling rate held fixed.	Recording-condition baseline; scoring within sampling-rate families.
In 22% of clips the signal band lies entirely above 8 kHz; 98% of mysticete clips were sampled below 16 kHz.	Analyse at the native sampling rate with settings in physical units.	Windows defined in seconds and hertz; unrecorded bands marked as missing, not as silence.
Three signal types exceeded the baseline across sources within the 40-kHz family (0.48–0.69 against 0.34).	Use coarse signal types as the first level of output.	Tonal, impulsive, pulsed, decided from tonal concentration and impulsiveness.
Finer categories were separated by single quantities (AUROC 0.95, 0.90, 0.76) but not beyond the baseline by any probe.	Report fine distinctions as measured values, not as classes.	Centre frequency, bandwidth, duration, frequency modulation, inter-click interval per event.
Above 8 kHz clips were 98–100% odontocete; below 1 kHz tonal and pulsed clips were 85–87% mysticete.	Give taxon as a probability conditioned on the band.	Look-up of taxon composition by signal type and band.
Between 1 and 8 kHz tonal clips were 51% odontocete, 26% mysticete and 24% pinniped.	Concentrate new feature development on the overlapping band.	Contour shape, harmonic spacing and pulse-interval regularity for 1–8 kHz signals.
39% of clips name more than one call type; song behaved unlike single sounds.	Describe sound events, not clips; allow overlap; treat song as a sequence.	Event detection before description; a clip or recording as a set of events.
Density-based clustering found two or three clusters in most representations; the feature space is continuous.	Prefer continuous values and probabilities to hard categories.	Outputs with uncertainty; rule-based names as a reading aid only.

4.6 Limitations and next steps

The results of Sections 3.5–3.7 are exploratory. The grouping into three signal types was chosen after the confusion among the eight categories had been seen, the exclusion of song and bark was decided at the same time, and no untouched data remain to confirm them: the held-out sources have been inspected. The pulsed type is small (197 clips) and mixes two categories that may not belong together. The neighbour and projection analyses used one random seed and carry no confidence intervals, and Table 3 covers the development sources only. The recording source is a proxy built from metadata. Detection in continuous recordings and non-biological sounds were not addressed, and the Watkins clips, with a median duration of 1.9 s, are not continuous recordings.

Three steps follow. The signal-type definitions and the descriptor thresholds should be fixed and verified by listening on a gold-standard subset. The three-stage method should be implemented and run on continuous recordings from an archive other than Watkins, where the grouping can be confirmed on data not used to choose it. And the 2,999 clips excluded or set aside here, most of them recorded in captivity, should be embedded to show where they fall relative to the eligible pool.

References

Au, W. W. L., & Hastings, M. C. (2008). *Principles of marine bioacoustics*. Springer.

Cominelli, S., Bellin, N., Brown, C. D., Rossi, V., & Lawson, J. (2024). Acoustic features as a tool to visualize and explore marine soundscapes: Applications illustrated using marine mammal passive acoustic monitoring datasets. *Ecology and Evolution*, 14(2), e10951. https://doi.org/10.1002/ece3.10951

Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2, 665–673. https://doi.org/10.1038/s42256-020-00257-z

Gillespie, D., Mellinger, D. K., Gordon, J., McLaren, D., Redmond, P., McHugh, R., Trinder, P., Deng, X.-Y., & Thode, A. (2008). PAMGUARD: Semiautomated, open source software for real-time acoustic detection and localisation of cetaceans. *Proceedings of the Institute of Acoustics*, 30(5).

Gong, Y., Chung, Y.-A., & Glass, J. (2021). AST: Audio Spectrogram Transformer. *Proceedings of Interspeech 2021*, 571–575. https://doi.org/10.21437/Interspeech.2021-698

Hershey, S., Chaudhuri, S., Ellis, D. P. W., Gemmeke, J. F., Jansen, A., Moore, R. C., Plakal, M., Platt, D., Saurous, R. A., Seybold, B., Slaney, M., Weiss, R. J., & Wilson, K. (2017). CNN architectures for large-scale audio classification. *Proceedings of ICASSP 2017*, 131–135. https://doi.org/10.1109/ICASSP.2017.7952132

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804. https://doi.org/10.1016/j.patter.2023.100804

Kim, M. J., Lee, J., Cho, Y., Kim, W.-K., Park, J., Lee, D., & Bae, H. S. (2026). Automated detection and classification of marine species vocalizations using a YOLO-based deep learning framework. *Ecology and Evolution*, 16(4), e73466. https://doi.org/10.1002/ece3.73466

Liang, et al. (2025). Automatic detection and unsupervised clustering-based classification of cetacean vocal signals. *Applied Sciences*, 15(7), 3585. https://doi.org/10.3390/app15073585

Lipari, V., Brancatelli, G., & Forlin, E. (2026). Marine mammal acoustic species recognition via metric embedding learning with Audio Spectrogram Transformer. *IEEE Access* (in press). https://doi.org/10.1109/ACCESS.2026.3714099

McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. arXiv:1802.03426.

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. https://doi.org/10.1111/ecog.02881

Sayigh, L., Daher, M. A., Allen, J., Gordon, H., Joyce, K., Stuhlmann, C., & Tyack, P. (2016). The Watkins Marine Mammal Sound Database: An online, freely accessible resource. *Proceedings of Meetings on Acoustics*, 27, 040013.

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proceedings of ICML 2019*, PMLR 97, 6105–6114.